

Team UpembaPark

Data Card template

Intelligent Complaint Classification via Transformer Architectures

Dataset: ComplaintSense Consumer Complaint Classification Challenge

Input vs. Output	• What are users inputting (e.g., audio, text, images)? • What are the outputs or labels your model will predict or generate?	Your response: A predicted complaint category from one of 10 predefined classes: <code>account_access</code>, <code>billing</code>, <code>customer_service</code>, <code>delivery_shipping</code>, <code>fraud_unauthorized</code>, <code>general_inquiry</code>, <code>product_defect</code>, <code>refund_return</code>, <code>subscription_cancel</code>, and <code>warranty_repair</code>.
Classification problem	• What is the classification task, if any? • What categories or classes are you working with, and why?	Your response: This is a supervised multi-class text classification problem. A complaint narrative is provided as input, and the model predicts the most appropriate complaint category. The dataset contains 10 predefined categories covering common consumer issues, including account access, billing, customer service, delivery and shipping, fraud or unauthorized activity, product defects, refunds and returns, subscription cancellation, and warranty or repair issues. These categories were provided by the competition and were used as the target labels without modification.
Labeling	• Where do your labels come from — domain experts, community members, crowd workers, AI-generated sources? • How will you check for consistency and accuracy?	Your response: Labels come from the competition-provided training dataset and were not generated by the project team or by an AI system. Before training, the data was checked for missing labels, duplicate records, class distribution, and consistency of the category values. The predefined labels were retained as supplied so that the model could be evaluated against the competition's intended classification task.
Data creation	• How will you collect your data?	Your response: The project uses the dataset supplied for the ComplaintSense Consumer Complaint

		Classification Challenge. The dataset contains 380 labeled training complaints and 160 unlabeled test complaints across 10 categories. Each training record contains a complaint identifier, complaint text, target Category, and FamilyId. FamilyId identifies complaints derived from the same underlying scenario and was used when creating the validation split to reduce the risk of related complaint variants appearing in both training and validation data.
Web collection ethics	If collecting from the web, how will you address consent, copyright, privacy, and legal/ethical considerations?	Your response: The project does not independently scrape or collect complaint narratives from the web. It uses the dataset supplied through the competition. The team does not attempt to identify individuals represented in the complaints, supplement the dataset with personal information, or use the data for purposes outside the classification task. Complaint narratives are treated as potentially sensitive text, and outputs are intended for research and educational purposes rather than decisions about individual consumers.
Bias and representation	- What groups may be underrepresented or excluded? - How will you improve diversity, fairness, and representation?	Your response: The dataset represents the complaints included in the competition and may not capture the full range of consumer experiences. Some types of complaints may be more frequent than others, while certain consumer experiences or complaint patterns may be underrepresented. The relatively small training set (380 labeled examples) also limits how broadly the model can generalize. To address these limitations, the project examines class distribution and evaluates performance using Balanced F1 (macro F1) rather than accuracy alone, giving each category equal importance. A family-aware validation split was also used to reduce leakage between related complaint examples.
Values and community alignment	- How does your dataset design reflect the values from your Challenge 1 problem statement? - How will it benefit and remain accountable to the intended community?	Your response: Privacy: The project avoids adding personal information or attempting to identify individuals from complaint narratives.

		Fairness: Model performance is evaluated across complaint categories rather than relying only on overall performance. Balanced F1 is used to reduce the influence of differences in category frequency. Human oversight: The model is designed to assist with organizing and classifying complaints. It is not intended to independently make high-stakes decisions about consumers. Accountability: Model limitations, validation performance, and classification errors are documented. In particular, the project recognizes that the model performs unevenly across categories and should therefore be treated as an assistive research system rather than a fully autonomous complaint-handling system. Human oversight: model assists classification/organization only — not an independent high-stakes decision-maker.
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